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Review

Leveraging Digital Twin Technology for Sustainable and Efficient Public Transportation

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Abstract: Public transportation systems face numerous challenges like traffic congestion, inconsistent schedules, and variable passenger demand. These issues lead to delays, overcrowding, and reduced patron satisfaction. Digital twin (DT) technology is a promising innovation for improving public transportation systems by offering real-time virtual representations of physical systems. By integrating real-time data from various sources, digital twins can enable predictive analytics, optimize operations, and improve the overall performance of public transportation networks. This work explores the potential of digital twins to optimize operational efficiency, enhance passenger experiences, and support sustainable urban mobility. A comprehensive review of the existing literature was conducted by analyzing case studies, theoretical models, and practical implementations to assess the effectiveness of DTs in transit systems. While the benefits of DTs are significant, their successful implementation in bus transportation systems is impeded by several challenges like scalability limitations, interoperability issues, and technical complexities involving data integration and IT infrastructure. This paper discusses ways to overcome these challenges, which include using modular designs, microservices, blockchain for security, and standardized communication for better integration. It emphasizes the importance of collaboration in research and practice to effectively apply digital twin technology to public transit systems.

Keywords: digital twin; public transportation; sustainable mobility; smart cities

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1. Introduction

1.1. Background and Context

Public transportation has been an integral part of urban mobility since the 19th century. It has also gone through many technological advancements to meet the growing needs of cities. Forms of public transportation systems in the United States include buses, trains, ferries, and airlines, which provide accessible and publicly financed services to millions of people. According to the American Public Transportation Association (APTA), public transit ridership grew by roughly 21% between 1997 and 2019, even as private vehicle ownership increased, highlighting its critical role in the U.S. economy. As of April 2024, APTA reported that public transit ridership, which had dropped to 20% of pre-pandemic levels in April 2020, had rebounded to 79% of pre-pandemic levels [1,2]. In terms of the highest public transit usage, New York, followed by Boston, Chicago, and Washington, DC, has the highest public transit ridership. Figure 1 presents a bar chart showing the most recently published paid passenger ridership (in millions) data in the four cities.

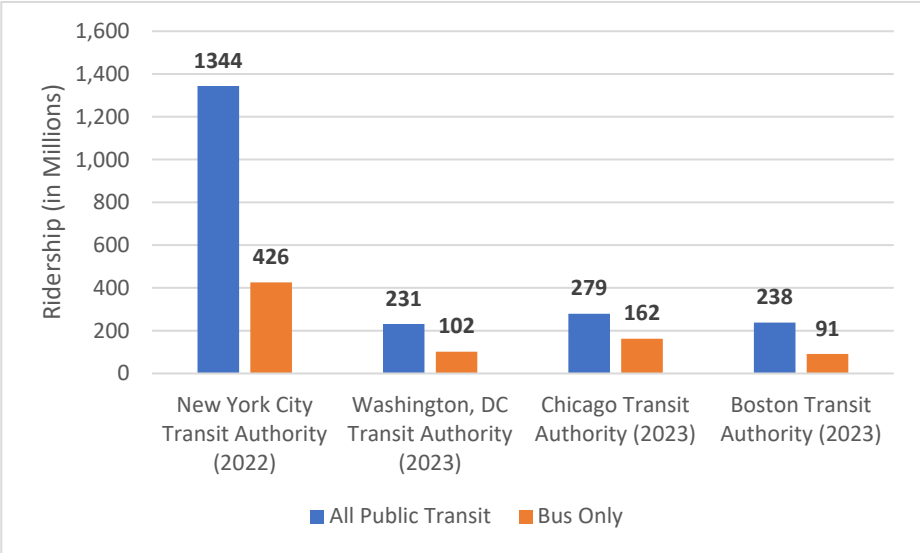

Figure 1. Transit ridership in US cities with the highest public transit usage [3–5].

In a global context, the most recent comprehensive public transportation ridership data are unavailable due to variations in data collection methods and inconsistent reporting standards. However, Figure 2 provides the bus rapid transit ridership details from 2022 for cities with the highest utilization of bus rapid transit systems. The numbers highlight the critical role of daily bus transit in urban mobility worldwide [6].

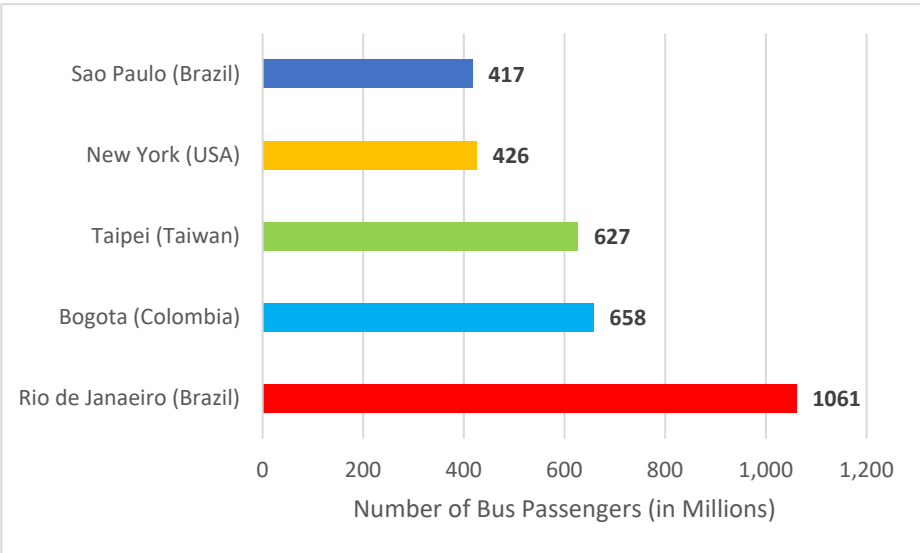

Figure 2. Bus ridership in cities with the highest bus rapid transit usage [6].

Urban transit agencies work continuously to improve operations, addressing challenges daily that include traffic congestion, unpredictable schedules, fluctuating passenger demand, and maintenance issues. These challenges grow with the rapid growth of urban cities and hence require more efficient, reliable, and sustainable solutions to keep up with the growth. Over the years, the traditional public transportation systems have operated on tight schedules and outdated infrastructure. As such, they lack the adaptability needed to respond swiftly and dynamically to unexpected disruptions in real time [7,8].

Sustainability is at the forefront of smart city initiatives. With rapid urbanization, the importance of efficient public transportation is further amplified as cities seek to reduce congestion, lower emissions, and enhance the quality of urban life. This can be achieved

through greener, more resilient transit solutions. Effective transit systems have the potential of reducing traffic congestion, lowering pollution levels, and enhancing overall quality of life. While advances in Intelligent Transportation Systems (ITSs) have improved urban traffic operations, many networks still rely on fixed schedules, unable to adjust to real-time events such as accidents, delays, or fluctuating demand on the road [9].

In recent years, the emergence of digital twin (DT) technology has offered a novel solution to the time constraint issues in urban cities. A digital twin is a virtual replica of a physical system, which can integrate real-time data to simulate, predict, and optimize performance [10]. With the help of this tool, it is possible to model and simulate real-world systems in various sectors such as healthcare, manufacturing, and smart cities [10]. In the realm of public transportation, digital twins have the capability of modeling entire networks and optimizing routes, vehicle maintenance, and passenger flows based on live conditions. Such an approach and dynamic integration could dramatically improve efficiency, reliability, and the user experience in urban public transit.

1.2. Research Problem and Gap

Despite various technological advancements, public transportation systems often struggle with real-time challenges such as traffic congestion, vehicle malfunctions, and fluctuating passenger demand [11]. These challenges lead to delays, overcrowding, higher operational costs, and reduced user satisfaction. Most traditional public transportation systems operate on outdated schedules and infrastructure. These factors limit the systems' ability to adapt dynamically to disruptions or optimize operations [12,13].

Digital twins can use real-time data from GPS, sensors, and passenger feedback, and they provide feedback to enable planners to predict issues like vehicle breakdowns or overcrowding [13]. Planners can also optimize operations and simulate future scenarios. Furthermore, there are also possibilities of adding new routes or adjusting schedules in real time. This adaptability can help improve decision-making, enhance the passenger experience, and make public transit systems more resilient to urban challenges. This can also help alleviate environmental concerns.

Although digital twin technology has been successful in many fields like manufacturing and smart city infrastructure [14], its potential for public transportation has not been fully explored. While there has been a lot of research focusing on traffic management and smart cities, more comprehensive and integrated public transit systems have not been studied in detail. Hence, there are limited studies that investigate digital twins and their usage of real-time data, advanced simulations, and predictive analytics to enhance urban public transit networks. As urban mobility is becoming more complex with a growing demand for efficient, data-driven solutions, it is important to explore how digital twins can improve public transportation through real-time data, simulations, and predictive analytics. This literature review aims to bridge this gap by compiling and analyzing existing knowledge to support the effective adoption of digital twins in public transit networks. Other initiatives include further identification of key challenges and outlining future research directions.

This research will explore how digital twin technology can optimize multimodal public transportation systems. The paper will identify challenges in integrating data sources such as real-time GPS, traffic conditions, vehicle diagnostics, and passenger feedback into a singular digital twin model, which can predict and mitigate issues in real time. Addressing this gap can produce efficient and adaptive public transit systems.

1.3. Objectives

The primary objective of this survey paper is to examine the potential of digital twin technology in improving the overall efficiency, reliability, and performance of public transportation systems. The key goals are the following:

1. Review the existing literature on system model development for digital twin integration in public transportation.
2. Identify key data sources for real-time monitoring and predictive analytics.
3. Analyze case studies and best practices in the use of digital twins for transit optimization.
4. Recommend future research directions and practical applications for digital twin technology in urban public transit systems.

1.4. Abbreviations and Literature Overview

Table 1 presents the abbreviated terms used in the paper.

Table 1. Description of abbreviated terms.

Abbreviations	Descriptions
Digital Twin	DT
American Public Transportation Association	APTA
Intelligent Transportation System	ITS
Adaptive Traffic Signal Control	ATSC
Autonomous Vehicle	AV
Electric Vehicle	EV
Artificial Intelligence	AI
Machine Learning	ML
Digital Twin Network	DTN
Radio Frequency Identification	RFID
Internet of Things	IoT
Global Positioning System	GPS
Automated Guided Vehicle	AGV
Internet of Vehicles	IoV
Vehicular Ad Hoc Network	VANET
Mean Absolute Percentage Error	MAPE
Root Mean Squared Error	RMSE
Artificial Neural Network	ANN
Mean Squared Error	MSE
Vehicle to Vehicle	V2V
Vehicle to Cloud	V2C
Vehicle to Pedestrians	V2P
Vehicle to Infrastructure	V2I
Traffic Accident	TA
Long Short-Term Memory	LSTM
Convolutional Neural Network	CNN
Asset Administration Shell	AAS
General Data Protection Regulation	GDPR
Department of Transportation	DOT

For this review, relevant papers and articles were sourced from trusted scholarly literature platforms like Google Scholar, ResearchGate, and Scopus. The quality of the papers was filtered by identifying reputable journal publications, peer-reviewed papers, conferences, seminars, as well as articles and books from reputable publishers like IEEE, Springer, MDPI, Taylor & Francis, and Elsevier, amongst others. Specific and relevant keywords like “Digital Twin”, “Artificial Intelligence”, “Public Transportation”, “Smart Cities”, “Machine Learning”, “Traffic Data”, and “Simulation” were used with the operator

AND. To further understand the major focus of these papers, the themes of the publications were categorized into six broad topics. A total of 97 papers were reviewed to prepare the Relevant Literature section.

Table 2 presents the bibliography of the reviewed papers, covering various aspects of digital twins, including their challenges, limitations, and real-world applications based on the identified six themes. The methodology for extracting relevant information for this review was based on the themes presented in this table. The "Citation Number" reflects the studies that have been cited in the References section.

Table 2. Annotated bibliography.

No.	Theme of Paper	Citation Number	Summary
1	General Digital Twin Application	[15–35]	The studies in this category comprehensively explore various aspects of general DT applications [15,17,22,24,28,30,31,33,34]. A few papers discuss theoretical advancements in DT technology, while others offer practical case studies with real-world implementations [16,18–21,25,35,36] Some studies examine the future potential of DTs in revolutionizing transportation, addressing challenges such as scalability, policy adaptation, and interdisciplinary collaboration [23,27,29,32].
2	Technology: Intelligent Transportation Systems, Blockchain, Internet of Things	[19,22,28,37–49]	The papers explore the integration of DTs with ITSs, blockchain, and the IoT, highlighting their role in improving efficiency, security, and decision-making [42,46,48]. Some studies discuss the deployment of IoT sensors for real-time traffic monitoring and V2V and V2X communications [19,28,38,39,41,44,45]. Studies also discuss blockchain's potential to enhance data security and ensure transparency in transportation systems [22,47]. Additionally, studies investigate how machine learning models and IoV use DTs to optimize vehicle routing, for predictive maintenance, and to reduce congestion [19,37,40,43,49].
3	Smart City and Sustainability	[17,19,50–63]	Studies highlight the transformative impact of DTs in advancing smart cities and sustainable urban development [19,50,53–56,61]. Many studies focus on the role of DTs in optimizing urban mobility, energy-efficient transportation, and dynamic traffic control [17,52,57,63–65]. Other studies discuss DTs in urban planning and the simulation and modeling of various infrastructural projects before implementation [51,58]. Studies also explore the use of green technologies, such as electric and autonomous public transportation, integrated with DT models for sustainable city growth [59,62,66].

Table 2. *Cont.*

No.	Theme of Paper	Citation Number	Summary
4	Traffic Data Management, Vehicular Network and Architecture	[28,48–50,57,60,66–80]	Research studies explore the application of DTs in traffic data management, vehicular networks, and transportation infrastructural architecture [28,48,60]. Several studies discuss data-driven approaches for real-time traffic optimization, using DT models to analyze congestion patterns and predict traffic flows [49,50,66,68–70,73–76,80]. Other studies highlight the significance of vehicular networks for V2X communication and intelligent transport protocols that enhance road safety and efficiency [71,72,77–79]. Studies also investigate the challenges of integrating DT technology into the existing transportation architecture, including scalability, interoperability, and real-time data-processing [28,48,57,67].
5	Predictive Analytics and Simulation	[17,18,34–36,49,54,55,72,81–102]	Studies discuss DTs in predictive analytics and simulation techniques in public transportation and urban mobility [18,34,72,85,94,99]. Some studies explore the use of DT models for demand forecasting and to optimize fleet management and help planners enhance scheduling efficiency [36,49,82,84,86,88,89]. Other studies develop simulation-based methodologies to test infrastructural projects for risk-free experimentation and validation of new transit policies before implementation [17,54,55,87,95,96,101,102]. Papers also focus on the predictive capabilities of DTs in reducing operational uncertainties, estimating vehicle performance, and improving the overall passenger experience [35,82,83,90–93,97,98,100].
6	Resilience and Risk Management	[24,31,59,60,79,99,103–111]	Studies analyze how DT technology can enhance the resilience and risk management of transportation systems, especially in response to disruptions, cyber threats, and infrastructural failures [59,103,104,107,108]. Studies explore DT-based risk assessment models that simulate natural disasters, accidents, and cyberattacks to develop mitigation strategies [31,99,106,110]. Other works investigate the role of cybersecurity in protecting transportation networks, emphasizing encryption techniques, secure data exchange protocols, and real-time anomaly detection [60,79,105,109]. Furthermore, studies also discuss how DTs can support post-incident recovery, helping decision-makers restore transportation services efficiently when there are disruptions [24,111].

2. Review of Different Aspects of Digital Twins

2.1. Digital Twins in Urban Infrastructure

A digital twin is a virtual representation of a physical system or process that integrates data in real time, using simulations and applying artificial intelligence to optimize performance [15]. The concept was introduced in the National Aeronautics and Space Administration (NASA) in the 1960s where the scientists developed a "living model" for conducting Apollo missions [16]. Their version of the digital twin was used in processing data to model events, aid in forensic analysis, and support future decision-making.

Recently, the digital twin has gained popularity in many areas of research since it can work as a virtual mirror of a physical system. This enables real-time synchronization with sensor data from the field (real world). This synchronization allows better decision-making and systems to carry out necessary actions in an optimized manner [17]. Hence, while working with a huge amount of data that are continuously generated within automated systems, digital twins enhance productivity by improving quality objectives and thus enable further automation [18].

The role of a digital twin in the age of smart cities is paramount, with data gathered through various electronic methods and sensors. A smart city may be viewed as using information and communication technologies to achieve better public services, based on comfort, maintenance, and sustainability concerns. By integrating the Internet of Things, 5G wireless systems, blockchain, co-computing, simulation, and artificial intelligence, a digital twin could revolutionize urban governance. In urban transportation, for example, there has been a revolutionary turn of events due to rapid advancements in the aforementioned technologies. This interconnectedness has allowed a digital twin to virtually simulate the interactions between people, vehicles, roads, and environments. Digital twins enabled by high-speed network capabilities enable real-time communication among moving objects—vehicles and infrastructure—to enable higher functions such as autonomous driving, vehicle platooning, and remote driving.

One of the ways that digital twins can analyze traffic conditions is by fusing 3D models of transportation infrastructure with real-time operational data. These models analyze data by utilizing AI and big data to visualize dynamic transportation scenarios. These virtual models contain realistic components such as roads, greenery, and vehicles, which create an immersive experience of urban traffic. Among various investigations on IoV, one has focused on digital twins, showing they can effectively optimize IoV data-sharing or increase the resiliency of the transportation network against disruptions. The researchers reported a system reduction in communication overheads (additional time, resources, or computational effort for managing and coordinating data transfer between different systems or components in a network) by upwards of 50%. Furthermore, they also reported improvement in the operational efficiency by almost 20% compared to traditional algorithms [19].

Digital twins can be applied at different levels of urban planning. One level is strategic planning for critical infrastructures like mobility systems and energy distribution. Another level of operational planning is for managing services, local mobility, and building performance. At the emergency level, digital twins can also be used to integrate resilience and sustainability for disaster preparedness. In smart city planning, digital twins can allow stakeholders to evaluate the impacts of proposed changes before they are implemented [17,18]. Digital twins link physical environments with digital infrastructures through sensors, wireless networks, and cloud computing, handling a great volume of data. The data go through sophisticated models that run simulated scenarios, options for evaluation, and strengths–weaknesses before actual decisions are made. In addition, the digital twin allows for interacting in real time with the users; hence, it opens up the possibility of directly controlling city operations [19].

Furthermore, digital twins have also been used in Adaptive Traffic Signal Control (ATSC). One study developed a digital twin-based ATSC system that reduces and redistributes the waiting times of vehicles at intersections [50]. The method considers waiting times at both the target and upstream intersections. This can be useful in optimizing the traffic flow in congested networks. It is also scalable to city-wide applications, going a long way in improving the driving experience for urban commuters.

2.2. Public Transportation and Digital Technologies

Recent advancements in digital twin technology have significantly impacted public transportation systems, offering promising improvements in efficiency, safety, and operational management. By using simulation platforms enriched with real-world data, autonomous vehicles (AVs) can be tested and trained in a variety of scenarios, which enhances decision-making and safety and accelerates the deployment of self-driving technologies. One study highlights that intelligent expressways, AVs, and ITSs remain the primary focus areas for the application of digital twins in traffic management. While progress has been made with data mining, cloud computing, and other data-processing technologies, the ability to handle and extract useful information from the massive volumes of transportation data still requires further development [20].

In urban settings, digital twins are central to the operation of intelligent traffic management systems [81]. Urban traffic management can benefit from digital twins that model entire urban networks intersection by intersection. By simulating different traffic scenarios, digital twins can analyze road network density and optimize solutions for traffic flow. This ensures more effective management of intersections equipped with numerous traffic sensors. With such systems in place, urban traffic control can automatically adjust and optimize based on real-time data, leading to improved road efficiency, safety, and efficient resource management.

In addition to urban streets, digital twins play a key role in the management of smart highways. Intelligent highway systems, which focus on areas such as toll collection, inspection, road condition monitoring, and all-weather access, leverage digital twin technology to process real-time data from vehicle and road sensors. This information is displayed to drivers via high-precision lane-level maps, assisting them in understanding current road conditions and the movement of surrounding vehicles. Moreover, these data are uploaded to a central digital twin platform, enabling traffic managers to issue early warnings and make informed decisions on road safety.

The use of digital twin technology in public transportation has the potential to revolutionize the sector [82]. By providing a digital identity for transportation assets, synchronized visualization, and real-time interaction between virtual and physical environments, digital twins enable traffic managers to control traffic perception, determine travel demand, issue road warnings, and coordinate emergency responses. These capabilities improve both safety and operational efficiency, offering new possibilities for intelligent driving and traffic management. A visual representation of a smart transportation platform based on digital twins is presented in Figure 3.

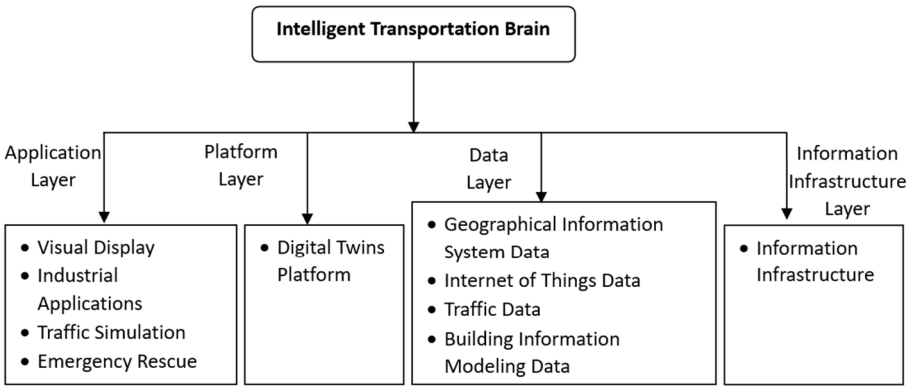

Figure 3. Representation of a smart transportation platform [82].

AVs (autonomous vehicles) and EVs (electric vehicles) have been the focus of recent studies, particularly digital twins, which are used for the development and testing of

electric propulsion systems. Regularly powered by electricity, these cars typically have advanced AI or machine learning (ML) technologies, which enable them to detect their environment, plan their routes, and drive safely [21]. Electric propulsion systems, which can be optimized using digital twin models running alongside ML algorithms, not only increase the efficiency of the vehicle but also improve the vehicle's performance. Digital twin networks (DTNs) are also part of the city's public transportation, and they offer updates like the real-time car status and vehicle safety information throughout the city. Therefore, the digital twin network plays a significant part in the development of the whole EV charging infrastructure scenario, as the city mobility undergoes significant changes. The same technology also allows for better understanding and controlling EV public transportation through simulation and feasibility studies as well.

The integration of ITSs with digital twins is yet another significant area of research. One research study proposed a system that would combine the use of sensing devices on the vehicle, real-time simulation models, and predictive maintenance technologies to track the conditions of different vehicle components in terms of their remaining useful life [64]. Digital twin technology stands as a landmark opportunity for cities and vehicles to interact, which will improve urban mobility and vehicle performance, thereby enhancing transport management [67]. As the system evolves, it is a step towards an integrated system that can move from a Digital Shadow and become a fully functional digital twin.

Blockchain technology, when combined with digital twins, opens up possibilities for the next generation of transport, and thus better service [22]. Guaranteeing secure data-sharing among transportation stakeholders, such as operators, regulators, and users, is among the benefits of blockchain's decentralized structure. The interaction between blockchain and digital twins can enhance safety, support sustainable transportation solutions, and optimize the available set of solutions.

2.3. Current Applications of Digital Twins in Public Bus Transportation

2.3.1. Fleet Management

Bus fleet management, when originally termed in the 1970s, primarily focused on the maintenance of fleets. Currently, fleet management is associated more with efficient routing and scheduling (operational), location tracking and accident detection (safety), and real-time decision-making communication to fleet operators. It is also concerned with asset management (energy efficiency), a longer and reliable service life, and overall sustainability [83]. Fleet management heavily relied on manual record-keeping and radio communication. Telematics systems and Global Positioning Systems were responsible for providing the precise locations of buses. Fleet management software was introduced to schedule maintenance, keep track of fuel consumption, and manage driver logs. More recently, Radio Frequency Identification (RFID) and the Internet of Things (IoT) have been implemented, whereby real-time tracking, inventory management, and decision-making have become more successful when using advanced data analytics [84].

Digital twins have also been used to improve fleet operations by prioritizing predictive maintenance, fuel efficiency, and asset management. For instance, one study noted the integration of high-fidelity simulations with on-board health management systems, maintenance history, and historical data of vehicles to point out the benefits of DTs [10]. The authors use multidisciplinary physics-based simulations to develop models and methodologies that are essential for certification and ensuring the life-cycle sustainability of vehicles. Concepts like real-time prediction of composite structures, sensitivity to microcracks, and dielectric spectroscopy were studied to measure material change properties and determine the durability of materials and predict degradation and failure [85].

There was also a study conducted to reduce the operation and maintenance costs of fleets by improving electrified bus energy efficiency at the fleet level through digitalization. The optimal operation performance for all bus routes was determined via dynamic programming. The goal was to improve energy efficiency and management by processing large volumes of data produced by bus fleets. Moreover, simulation including real-world conditions such as traffic congestion, passenger variations, and auxiliary loads (air conditioning and light usage, driving motors) was also performed to determine the optimal operation, resulting in efficient fuel consumption by using dynamic programming. Fuel and grid cost calculations were evaluated to develop an economic model for fleet learning. Lower fuel consumption was observed, with fleet energy efficiency increasing by up to 46%, and fuel consumption decreasing by almost 49% when compared to historical data [37].

Another paper evaluated the electrification of public transportation as a sustainable mobility solution to address environmental concerns and improve public transportation systems. Raising concerns of trust and security, a blockchain-based system was proposed for fleet management and managing the charging of electric buses to address these challenges. Timely fleet schedules are proposed to be conducted at the least charging costs by analyzing the Quito-Ecuador bus fleet charging [86]. In a study conducted to determine the usage of alternative operation of fleets, researchers assessed future scenarios of municipal vehicles composed of alternatively powered vehicle types compared to the traditional fleet. It was found that the change to alternative fuel vehicles (battery and fuel cells) requires changes in the existing infrastructure. By analyzing existing conditions for fleet composition and infrastructure, the authors pointed out challenges such as developing new refueling concepts and facilities to accommodate different types of fuels. However, the authors pointed out that these complex logistical and financial undertakings can be assessed and simulated by developing digital twins. Future scenarios involving fleets consisting of alternatively powered vehicles, as well as strategic planning and efficient decision-making to optimize fleet performance, can lead to overall operational costs and sustainable choices [87].

Enhancing operational efficiency and improving the safety and monitoring of bus operation through the integration of digitalization in transportation systems was studied in Badalona, Spain. To understand the bus dynamics within the urban transportation network, a genetic algorithm was used to identify suitable configurations for simulating city traffic based on real data and bus schedule information. The authors were successful in developing a DT that is effective in mirroring the actual dynamics of the urban transportation system and can also adapt to possible anomalies [87]. A similar study was performed using the principle of a queuing system and modeling the concepts of a DT using Python programming language. The prediction of queue lengths and waiting times allows operators and management to prioritize vehicle fleet maintenance and repair activities in addition to resource planning and the fleet management structure. In the model, users can evaluate potential parameters of the enterprise (fleet) based on certain target indicator changes. It was concluded that the digital model serves as a great resource for forecasting activities, as well as developing business plans [88].

The authors of one study mentioned the concept of Automated Guided Vehicles (AGVs), where the research involved creating a virtual model of the physical AGV fleet in a controlled environment. The decision-making processes were improved by using optimization algorithms, and continuous improvement was achieved through feedback loops in the study. While the authors raised concerns about large-scale applications, they also concluded that extensive research should be conducted as DTs for AGVs have significant advantages over traditional fleet management techniques. The scalability in terms of learning-based resource management, particularly through deep reinforcement

learning, was also discussed in another study focusing on complex resource management using the Internet of Vehicles (IoV). However, it was noted that DTs integrated with AI-based resource management are promising for highly dynamic networks [89]. By providing a data-driven approach to decision-making, transportation planners and operators can better allocate resources and implement policies. This can lead to the overall enhancement of public transport systems and provide better experience for passengers and increased trust in public transportation services. Furthermore, the DT model's adaptability makes it scalable and flexible, allowing it to be applied to different cities and transportation systems.

2.3.2. Traffic and Route Optimization

Traffic congestion, especially on busy roads, can be greatly alleviated by optimizing bus routes. Furthermore, operators can prioritize efficient use of resources via data-driven decision-making that can not only improve passenger experience but also reduce emissions, leading to environmental benefits. One study to support optimized bus scheduling and passenger flow management discusses efficiently managing the number of vehicles to meet passenger demand while minimizing wait times for passengers. For instance, the algorithm proposed by the authors saw a reduction in the number of drivers required, leading to cost cutting [51]. Another study indicates that planners can make informed decisions when traffic optimization simulation models can predict the relationship between changes in bus routes and overall traffic patterns and congestion levels [90]. Improving traffic flow can also be achieved by optimizing traffic signal operations in urban areas. The authors of a study used decentralized graph-based multi-agent reinforcement learning incorporating DTs for traffic signal optimization. An approximately 55.38% reduction in fuel consumption over a 24-h scenario was achieved. The authors highlighted alleviating traffic congestion, increasing transportation efficiency, and enhancing the overall commuter experience as benefits in a larger-scale real-world application [68].

Research regarding Vehicular Ad Hoc Networks (VANETs) discusses challenges such as VANETs' frequent network disconnections, irregular connectivity, lack of node management, and seamless communication/collaboration between different vehicles, which limit proper traffic optimization. To overcome this, the authors incorporate DTs into the networks, and that resulted in an increased energy efficiency, network lifespan, and data transmission. They also highlighted that the combined model was able to analyze network behavior and predict traffic patterns [65]. Urban traffic signal control using VANETs to improve traffic management has also been performed using multimodal traffic simulation software SUMO. The authors used a real-time vehicle fuel consumption index while modeling the DT via an optimization algorithm, iterative optimization, and green wave control simulations (coordinating phase differences between signal lights of the major road) [52].

The concept of DTs was also found to be useful for Demand-Responsive Transit (DRT) to better simulate the relationship of traveler trip chain with the bus operation chain. The authors were successfully able to describe and validate the bus transit process between two stations in Anhui Province, China by using Automatic Vehicle Identification data. It was found that although the two models (one with and one without DTs) took largely the same time between two stations, passengers in the model with DTs had more seats available, thus enhancing the travel process by adjusting real-time supply and demand [91]. Like DRT, a dynamic bus-routing approach has also been studied to prioritize flexible transportation systems and passenger convenience. Using AI, the authors also optimize the average travel time by finding a globally optimal set of paths for a bus fleet [53].

2.3.3. Passenger Experience

Intelligent demand prediction and rerouting models, as well as facility design factors to increase passenger comfort, have been proposed in multiple studies to improve patron satisfaction [23,53,87,88,91]. More specifically, in public transportation, the passenger experience and comfort quality is a significant metric used to determine the standard of public transportation. In smart city concepts, the simulation of not only routes and schedules that reduce wait times but also of different policy scenarios can enhance the passenger experience. The cohesiveness of different transportation modes in smart cities can reduce inconvenience for passengers if they must switch between different modes. Provision of real-time information in terms of schedules, probable delays, and alternative routes allows passengers to operate on dynamic individual preferences (personalized), ultimately improving transportation management [54,69].

A study focused on incorporating passenger feedback and participation when developing more user-centered DT models, promoting occupant interaction [70]. In one study conducted in Gujarat, India, passenger estimation at each stop was found to be very accurate (predicted value almost equal to the actual value for two routes on weekdays and weekends). This was highlighted by a low Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) [23]. In an experimental study conducted by collecting real-time five-level Likert-scale ratings by bus patrons, and evaluating external factors and indicators like whole-body vibration due to acceleration and jerk, ride comfort was assessed using ANN algorithms. The ANN model generated an average Mean Squared Error (MSE) of 0.03, indicating a low level of error in predictions. The model also demonstrated a strong correlation between the independent variables and the dependent variable (level of comfort as rated by the passengers), with an R-value of 0.83 [93].

2.3.4. Safety and Emergency Response

The concept of creating a virtual replica of a physical urban area and integrating data from multiple sensors and IoT devices in real time allows for monitoring the dynamic traffic conditions. Technologies such as sensors, 5G, and Wi-Fi enable data to be captured from physical spaces and shared with other vehicles (V2V), the cloud (V2C), pedestrians (V2P), as well as infrastructure (V2I). The high automation and intelligent control that can be achieved from planning to maintenance can encompass various transportation modes, including buses [38,71]. Thus, DTs can enhance real-time traffic management by simulating multiple scenarios which can help in collision prevention and predictive analysis. DTs support emergency management by offering a comprehensive visualization of urban ecosystems. This visualization can help authorities understand the spatial dynamics of emergencies, enabling better planning and response, ultimately enhancing real-life emergency preparedness [55]. The authors of one study highlight that real-time traffic signal adjustments and routing can minimize collision risks, assess potential impacts, and develop effective response strategies. Further, dynamic response planning can be effective in events like natural disasters, where evacuation routes and emergency responses to protect civil infrastructures are prioritized [24,56].

The concept of a transportation digital twin (TDT) system can also improve the safety and mobility of existing urban systems while reducing emissions. Not only can the route of emergency vehicles be optimized by optimizing physical traffic signals, but the systems can also be used to close and restrict certain routes in case of potential hazards, facilitating urban transportation management [72]. Although not mentioned explicitly, implementing real-time preventive measures and route optimizing can be very beneficial to avoid high-risk areas or times for public transportation vehicles [94]. Research was conducted on risk evaluation for a highway traffic management framework by integrating high-risk

roadway identification, risk judgment, and control. Since buses are a significant component of highway traffic, the neural network for road segment risk identification could be applied to assess and manage risks associated with transit operations, ensuring safer and more efficient bus transit on highways [39].

Ensuring the safety and efficiency of bus operations is critical for maintaining public trust and encouraging the use of public transport. A study focused on vehicle stability analysis, trajectory deviation analysis, and risk assessment of vehicles driving along an expressway. The authors then created a DT to monitor various indicators like speed, sideslip, and rollover virtually [95]. Model Predictive Control, where the shared steering controller allows for balancing multiple goals (safety, efficiency, passenger comfort), and DT-based trajectory prediction have also been studied to decrease the chances of collisions due to inconsistent human operations such as aggressive/emergency lane change. This can support bus drivers in making quick decisions, especially in high-traffic situations [73]. Another study developed a traffic accident (TA) prediction system that uses a data service interface to connect with the business database, allowing it to access and retrieve business information to create a geographic information access view. Long Short-Term Memory (LSTM) network application to predict traffic accident time-series data can also be used for bus transit systems. By examining historical data related to specific bus routes and schedules, transit authorities can forecast potential accident-prone locations and times, enabling them to implement proactive safety measures [96].

Technological advancements can aid in real-time safety enhancements such as object detection, morion tracking, spatial attention, etc. It was possible to apply an algorithm for low-speed collision prevention warning to a DT based on a Convolutional Neural Network (CNN). Based on the experimental results, the algorithm demonstrated an excellent detection performance in different scenarios, with detection precision rates of 90.6% to 95.7% depending on the roadway type [74]. Another study focused on differentiating normal and abnormal data (dangerous drivers), enabling the prediction of dangerous areas to help make early warning decisions. Research on driver risk-aware intelligent mobility analytics discusses using real-time data for not only traffic volume estimation for planners to optimize bus frequencies and schedules but also for proactive incident management, whereby authorities can respond quickly to unexpected events. The DT model achieved lower MAPE (8.40% to 15.11%) compared to the base model (40.63–43.94%) when just traffic data were used. A lower MAPE range of 0.85% to 12.97% was obtained when individual driver behaviors and driving risks in various traffic conditions were evaluated. These phenomena and concepts can be greatly leveraged by transit operators for smooth bus operation in urban areas [40,72].

3. Challenges and Limitations of Digital Twins

3.1. Scalability

Scalability is the ability of a particular system to function, remain responsive, and operate smoothly with an increase in the amount of work or data. The scalability of DTs is a primary concern when deploying technology, especially in bus transportation systems in urban areas. Many researchers have pointed out the lack of capacity to integrate more data from data sources than originally expected and handle increased computational demands for DT simulations to be performed in real case scenarios.

Network architecture and application/technology stacking are some of the primary considerations when it comes to the scalability of DTs. A study discusses interoperability (ability of different systems to work together), security, and the correct determination of information sources for the long-term deployment of digital twins [25]. It is essential in bus transportation as multiple data streams from various sensors and systems must be

integrated seamlessly in real time. Another study presented a methodology that prioritized digital twin use cases based on stakeholder satisfaction and infrastructure scalability, which is essential for effective implementation in transportation systems. The authors, however, mention a lack of standardized approach, which leads to inconsistencies and challenges in effectively managing and scaling DT applications across different industries and stakeholders [41].

DT technology also requires a systematic approach, which necessitates specific environmental needs. A study points out that there is a lack of a comprehensive methodology for developing scalable and reusable digital twin solutions tailored to manufacturing environments. This is concerning since the extrapolation to the public transportation needs to be smooth and flawless. The authors suggest the requirement of a broader framework that can readily adapt to unique challenges posed by bus transportation, such as real-time data-processing and dynamic routing [36]. Real-time data integration and processing, and bidirectional data interaction capabilities, are important for bus transportation. While the data integration for operational adjustments can be accurate, scalability is essential to improve efficiency and passenger services. The authors conclude that the management of challenges related to data management, model complexity, real-time integration, resource allocation, and adaptability is important to achieve scalability in energy consumption [57]. The concept of modularity (a system which can be broken down into separate parts that can also be combined to function as a whole unit) is also important to facilitate scalability. A study discusses a microservices-based architecture which focuses on reusability and modularity to enable dynamic responses to meet the growing demands in industrial settings. This approach can be beneficial in the context of bus transportation, where different operations require flexible and scalable solutions [42].

Furthermore, the integration of digital twins with emerging technologies, such as the Internet of Things (IoT), can enhance scalability. One study highlights the potential of the IoT and digital twins to create advanced features in smart grids, which can be analogous to the smart transportation systems that utilize digital twins for improved operational efficiency [97]. The synergy between these technologies can lead to more robust and scalable solutions in bus transportation. Additionally, combining DTs with the IoT can also be beneficial in increasing smart grid capabilities, which in turn can prove to be reliable for smart transportation systems. A multi-layered security strategy where traditional security techniques (such as encryption, firewalls, and access controls) are integrated with IoT-based security measures can prove to be beneficial for DT security and resiliency. The authors of one study highlight the combination of technologies that can result in reliable and more adaptable solutions for all forms of transportation, including bus transportation [103]. While scaling digital twins in bus transportation has challenges, there is also the potential to improve efficiency and services.

3.2. Interoperability

Interoperability refers to the connection between the physical element or the infrastructure and its digital counterpart. Successful implementation of DTs requires seamless data exchange and communication among various virtual technologies and their physical counterparts. Development of standardized communication protocols is required to have compatibility as well as efficient data exchange across different DTs.

One of the primary challenges in achieving interoperability is the need for standardized communication protocols and data formats. The authors of one study pointed out that connecting data from different digital twins can lead to compatibility problems. There is thus a need for standardization to ensure smooth data exchange [75]. Another study mentions that digital twins in logistics require strong IT systems and consistent data man-

agement to work well together [58]. Without these standards, digital twins cannot fully improve transportation systems to achieve maximum efficiency. One of the key tools identified by the authors of one study is the Asset Administration Shell (AAS), which they used as a common interface that helps different digital twins communicate. The AAS is a standardization of DTs for industrial applications, which allows cross-company interoperability across multiple platforms. Using the AAS, transportation stakeholders can ensure the interoperability of various DTs and their components, making transportation systems more efficient and reliable. This kind of integration is especially important in smart cities, where many digital twins need to work cohesively to manage urban mobility [76,104].

Synchronization of data between digital twins and their physical counterparts is also crucial. There is a need to keep DTs up to date with accurate data synchronization to avoid delays in interventions. This synchronization is more complicated when cybersecurity concerns arise. Hence, there is a need for secure and reliable communication channels to ensure the integrity of data exchanged between digital twins and their physical systems [43,77].

The architecture of digital twins also plays an important role in their interoperability. Authors discuss a 'mobility digital twin' that relies on a clear setup to combine data from different sources and allow these systems to communicate easily [98]. Since different transportation systems use diverse technologies and data formats, the architecture of digital twins needs to handle this variety. For instance, creating digital twins that can work together in bus systems requires a strong tech setup that can manage real-time data-sharing. This includes the use of advanced communication protocols and data models that enable different digital twins to interact seamlessly [44]. Modular software architecture, mentioned in recent studies, also helps digital twins work together better by allowing for quick setup and flexibility for changing needs [99]. This approach reduces the engineering work, making digital twins easier to implement for transportation agencies.

Furthermore, integration technologies, such as IoT and 5G, can enhance the interoperability of digital twins in transportation. While the authors of one study mainly discuss the IoT with digital twins in power systems, the principles of the IoT and digital twins can similarly be applied to transportation systems, facilitating real-time data exchange and improving responsiveness [60]. This integration is crucial for autonomous vehicles, where low-latency communication is required for decision-making processes [63]. If interoperability challenges are addressed, these technologies can significantly improve the operational efficiency of bus transportation systems.

3.3. Technical Complexities

Integration of various data sources and technologies is one of the primary technical challenges in implementing DTs. Since DTs rely on real-time data from sensors, IoT devices, to create reliable virtual representations of physical assets, successful integration requires a complex IT infrastructure that can handle large volumes of data. This should be achieved to ensure seamless communication between physical and digital systems [100]. Management and data are also integral components of DTs. Effective data analysis is important for optimizing bus operations and improving service delivery. Some studies indicate that many organizations face problems in data management, including data silos, inconsistent data formats, and a lack of standardized methodologies for data integration [98]. The reliability and effectiveness of transportation systems are compromised when issues like these arise.

The setting up and maintenance of the necessary IT infrastructure is also a challenge. The implementation of DTs can be a long and costly undertaking for many stakeholders since high-performance IT infrastructure is required for their operation [75]. Hardware and software components along with regular maintenance and updates are required to

ensure DTs can represent the physical systems accurately as well as have continuous data synchronization. There are high associated cost and resource constraints when high-performance computing, cloud storage, and real-time data-processing capabilities are required. Multiple hardware and software requirements to collect and transit data efficiently are also of utmost importance. Continuous complex data synchronization to allow real-time decision-making while avoiding latency issues is also crucial. The lack or inefficiency of the infrastructure can lead to poor data accuracy, increased maintenance costs, system downtimes, and overall operational inefficiencies in bus transportation systems [26]. This can potentially lead the DT framework to be unreliable and incorrectly predict congestion patterns, inefficient bus routing, and increased passenger wait times.

Technologies using DTs have different requirements for what level of detail is required to develop an accurate model of real-world conditions. Simulated environments used for training AVs require a high level of detail, like road surface simulation, realistic roads, realistic traffic flow, and sensor and accident simulation, which are necessary to properly train AV decision-making in a virtual environment [78]. The quality of the simulated data or environment has a direct impact on the decision-making of the AI system. Additionally, virtual environments may not be useful for training classification models operating on direct sensor data because of the small differences in simulations compared to the real world [27].

LOD choice plays a vital role in balancing accuracy, cost, and computational efficiency in DT applications. Higher LOD requirements generally lead to increased computational complexity, higher costs, and greater data storage and processing demands. However, they also provide accurate and reliable datasets that improve model performance. Conversely, lower LOD models are more cost-effective, easier to implement, and require less data. These factors are beneficial for applications where real-time processing, privacy, and security are priorities. For instance, a low LOD DT used for high-level traffic flow analysis may only require total vehicle counts and speed data, whereas a high LOD DT for AV training must incorporate intricate details (granular) such as lane markings, pedestrian behavior, and environmental conditions [27].

The cross application of DT technologies will apply unique issues. AVs and fleet and traffic management have unique tradeoffs, which enable the technologies to work in real time while being performant. Modern vehicles are trending towards predictive maintenance and already have a lot of stored operational data [45]. AVs enhance this trend and have dedicated hardware to compute and gather information on AI systems, allowing them to make decisions based on the road environment. Vehicles will need additional hardware and processes to interact with ITSs, to communicate the road conditions and accident factors or traffic status, which is likely to increase the capital cost of vehicles.

3.4. Security

In the case of DTs, vast amounts of data are generated for smooth operations. With a multitude of IoT devices interconnected in the smart transportation system, it becomes complicated to properly secure data-sharing frameworks [105]. A study discusses a model that prioritizes the importance of data abstraction and control. The authors believe that security vulnerabilities in connected transportation systems can be significantly mitigated. The authors of another study also point out and classify numerous security and privacy vulnerabilities in the ITS. The risk of malicious attacks is amplified with the increase in connectivity [106]. Since most DT applications are cloud-based and often vulnerable to cyberattacks, there is a need for strict security protocols [79].

Identifying vulnerabilities through digital twin simulations can help in planning effective countermeasures against potential cyberattacks. Secure frameworks are necessary

to manage data generated by IoT devices within smart ecosystems [59,79,105,106]. The integration of digital twins in transportation systems can help authorities identify and address potential problems, like security threats, before they occur in the physical world. For instance, the maritime transportation sector has already recognized the importance of securing IoT-enabled DTs, which can also be inferred for bus transportation systems. The insights from maritime transportation apply to bus transit systems, as both use the IoT for monitoring, route optimization, and maintenance. Like maritime DTs, bus systems need strong cybersecurity to prevent potential threats such as data breaches, unauthorized access, and cyberattacks [28,107]. Hence, robust security measures need to be prioritized for both the digital and physical realms of transportation systems.

To ensure data security for passenger and bus operational data, the authors propose that organizations can adopt several strategies like encryption, access controls, and threat detection mechanisms [46]. The combination of blockchain technology with DTs has also been proposed as a possible solution to enhance data security in smart transportation systems. By securely storing data related to traffic events and user interactions such as traffic events and user reputation, blockchain can mitigate risks associated with data breaches and unauthorized access. The authors of one study developed a software application using technologies such as libp2p (peer-to-peer networking), which contained functionalities for creating transactions and blocks to verify data integrity and displaying the blockchain. The aim was to make processes accessible, regardless of the number of layers required to be passed, ensuring the security of the new connection effortlessly [47].

3.5. Privacy

The implementation of DTs in public transportation must also consider the policies and concerns surrounding data privacy. The policies regarding data collection, usage, and sharing to build trust with users and stakeholders must be clearly upheld by organizations. One example of privacy and policy implication is that technology must be compliant with legal frameworks, such as the General Data Protection Regulation (GDPR), to ensure that user data are handled responsibly and ethically [112]. Communication methods in DT environments are also critical to ensure data privacy. One study presents a privacy-preserving communication scheme using blockchain technology to secure data exchanges in cloud-based DT environments. The authors highlight the effectiveness of blockchain in providing a decentralized approach to data security, which is applicable in bus transportation systems where sensitive passenger data are transmitted frequently and in real time [48]. The authors of another study also point out the need for robust security measures in 5G-integrated IoT environments by exploring privacy-preserving schemes in smart transportation [29].

A study provides a comprehensive survey of security threats associated with digital twin technologies by breaking down potential security threats to digital twins (DTs) into clear categories to understand the risks involved in using them. The authors emphasize the reliance on the combination of multiple technologies, and how the connection to the physical world can create unseen security challenges. By examining the holistic approach to security and operation and identifying possible weak points, the authors discuss potential vulnerabilities that could arise when deploying DTs [107]. In addition to technological solutions, organizational strategies play a crucial role in protecting data privacy. Organizations can implement cloud-based solutions to protect sensitive data as well as user privacy in intelligent transportation applications to address both technical and organizational challenges [108].

The authors of one study discussed the integration of advanced technologies such as 6G networks to enhance the performance and security of digital twin edge networks. However, these advancements also require a reassessment of existing security and privacy

measures to address the unique challenges posed by next-generation connectivity. The authors suggest a proactive approach to security that uses the capabilities of emerging technologies to create resilient transportation systems [30].

Furthermore, the role of user awareness and education in mitigating privacy and security risks is also essential to the smooth operation of DT technology. A well-informed and trained workforce is essential for managing security risks and implementing the best practices in DT systems. Training in cybersecurity, data privacy, and interoperability can strengthen transportation networks. Similarly, public awareness campaigns can enhance transit users' understanding of operations and importance of data security and allow continuous feedback to be gained, to improve overall DT systems. One study highlights the importance of user education in smart grid environments, which is also applicable to smart transportation systems [31]. The authors of another study discussed the importance of user engagement to develop secure IoT applications. This can foster a culture of security awareness among users, and organizations can ultimately enhance the overall security of their transportation systems as well [32].

3.6. High Costs and Infrastructural Requirements

High costs and infrastructural requirements are also technical challenges that can arise in the process of implementing DT technology. Digital twins can create better fleet management, optimize routing, and enhance maintenance schedules. This can be beneficial to agencies as well as the public transport used in the long run. However, the initial investment required can prevent many organizations from initiating DT technology. A study highlights energy efficiency and reduced operational costs of DT implementation but also mentions that the upfront financial commitment is an inconvenience [78]. Apart from initial setup costs to adopt DTs in bus transportation, ongoing maintenance and operational expenses should also be considered. Regular updates to not only the software but also hardware, along with personnel training/management, add to the overall operating costs [33]. In one study, the authors mention that state Departments of Transportation (DOTs) face significant challenges in managing multiple assets across extensive networks. Since integration requires investment in both technology and training, many transportation agencies can be discouraged [66].

Proper functioning of DTs requires a solid technological foundation that includes advanced sensors, data analytics, and communication systems all working with each other. The bus and transit infrastructure relies on the seamless integration of these components. Using Internet of Things (IoT) devices is essential for data collection and real-time monitoring of bus fleets, but this can be costly and require significant infrastructural upgrades. For example, using networks such as 5G/6G can increase the costs associated with implementing digital twins in transportation systems [33,60]. Successful implementation of DTs in bus transportation systems relies on managing and analyzing the large amounts of data produced in real time. Handling data from multiple sources in different formats requires advanced data analysis tools and strong systems to ensure the data are accurate and reliable. However, developing and maintaining these systems can be expensive, especially for smaller transit agencies [80]. The complexity of digital twins also requires a deep understanding of how transportation systems work as a whole. Without a clear framework to connect all parts of the transportation network, the potential benefits of digital twins might not be fully realized. As a result, transportation agencies require skilled professionals to manage and interpret the massive amounts of data generated by digital twins, adding another layer of difficulty to their implementation [61]

Some transit agencies that operate on limited budgets can consider these technological and skilled manpower requirements as financial burdens. Another concern is the high

computational cost associated with simulation and modeling of DT environments. The authors of one study discussed the impracticality of widespread DT application because of associated computational expenses. Another study indicates that the aging and deteriorating infrastructure can pose problems with effective management. Since the systems require up-to-date and real-time information to perform optimally, advanced computational resources are required for ideal operation conditions [101,109]. The need for extensive infrastructure includes not only the physical assets on its own but also the underlying data management systems for a DT framework. One study points out the multiple stages of DT implementation where the technology involved and its integration into the existing systems are carefully considered. The increase in the complexity of these systems will lead to increased operational costs to ensure that the organizational infrastructure can support the continuous data flow and processing demands of digital twins [110].

In conclusion, DT technology has the potential to improve bus transportation systems, but it also comes with significant challenges. Issues such as data privacy, security, high costs, and complex infrastructural requirements can demotivate and slow widespread adoption. Scalability, interoperability, and technical complexities associated with the implementation of digital twins in bus transportation systems include challenges like data integration, management, IT infrastructural requirements, and the need for standardized methodologies. To fully leverage the benefits of DTs to better transportation efficiency, organizations must implement robust security measures, ensure interoperability, adhere to regulatory requirements, and address the financial and operational barriers associated with their deployment.

4. Case Studies and Applications

Case studies for DT technological application were searched and filtered to incorporate diversity in technological and geographical aspects in addition to real-world applications. This was to ensure the inclusion of different operational aspects of bus transport systems, such as traffic flow management, maintenance optimization, and route planning from different regions. The implementation of digital twin (DT) technology in bus transport systems has demonstrated measurable improvements in various operational aspects, including reduced delays, enhanced maintenance cycles, and increased passenger satisfaction. These improvements result from DT's ability to provide real-time data analysis and predictive maintenance capabilities [53].

Reduction in delays is one of the main benefits of DTs when it comes to bus transport. By utilizing real-time data from sensors and IoT devices, digital twins can simulate and predict traffic patterns, enabling better route planning and scheduling. For instance, the authors of one study highlight the role of DTs in ITSs, which can facilitate real-time monitoring and management of traffic flows to minimize delays at intersections and along bus routes [110]. The authors implemented a DT at a ramp merge, non-signalized intersection in Brazil with the aim of building models and applications, which will allow visualize scenarios to analyze accident situations and avoid them. Additionally, the systematic mapping study mentions that DTs can diagnose and address issues in transportation systems, while suggesting that most existing approaches are limited to system monitoring or fault detection rather than comprehensive diagnostic reasoning [102].

By utilizing real-time data analytics and simulation capabilities, DTs can predict and mitigate potential delays caused by traffic congestion, which can be very helpful in bus transit systems as well. As detailed in Section 2.3 of this paper, the case study conducted in Badalona, Spain demonstrates how a DT enhances bus operations by improving transit efficiency and safety. The authors were able to utilize a genetic algorithm to model traffic and schedules, which adapts to anomalies, enabling real-time system monitoring [87].

In another study that evaluated a day-long simulation in Geneva, the model was able to accurately replicate the real traffic (where traffic counters were present) using SUMO [113]. In another study, by incorporating schedule recovery on a simulated bus route in Guangzhou, China, the authors were able to demonstrate a reduction in waiting times, as well as bus travel times. This indicates that integrating DT frameworks with traffic data can enhance the management of bus schedules if implemented properly by authorities to optimize routes and minimize wait times for passengers [49]. Improvement of the passenger experience was also discussed in Section 2.3, where a study compared two models, with and without a DT, in Anhui Province, China [91]. The direct benefits of incorporating a DT into the bus transit process model provided more seating availability for passengers to address supply and demand in transit. Furthermore, the application of such technologies allows for dynamic adjustments to bus frequencies based on real-time demand, which has been shown to improve efficiency, effectiveness, and scalability [111].

Digital twins also enhance the efficiency of maintenance operations by utilizing predictive analytics for maintenance requirements based on real-time performance and usage data. This proactive approach can help prevent unexpected failures, thus extending the lifespan of bus components and reducing operational costs. Studies have highlighted DTs' ability to optimize maintenance planning and minimize downtime by utilizing insights derived from comprehensive data analysis from various health monitoring sensors in buses [34,62,100] Furthermore, as mentioned in Section 2.3, passenger engagement was also explored in user-centered digital twin models in Gujarat, India [23]. The study integrated feedback to improve bus service reliability and real-time adaptability, which highlights the role of user-centered approaches in enhancing DT effectiveness.

A study in Mountain View, California testing personalized adaptive cruise control by applying a Mobility DT (MDT) based on an AI data-driven cloud-edge framework demonstrated an improvement in the level of driver comfort on the road by approximately 8.4% (events of drivers applying the brake or acceleration to take over the cruise control occurring 86.4% less frequently) in 2021. End-to-end testing demonstrates that the proposed MDT framework can efficiently handle complex cloud computing tasks using bulk data, with minimal delay. This improves the performance of vehicle applications. For real-time applications, the MDT framework demonstrated that the latency is under 80 milliseconds, which is crucial for safety and time-sensitive operations [35,63]. In another study, where authors implemented a DT for an electric vehicle to study the interaction with an urban DT environment, real-time and static data were collected from various sensors to analyze performance, energy usage, and sustainability. The study found that there was a distance estimation average percent error of only 1.8%, corresponding to 5 m. This indicates high accuracy in the mapping module, which can be very beneficial when implemented by the bus transit system for real-time route optimization. The authors also concluded that the incorporation of a LIDAR-IMU algorithm can further reduce the error to 1% (less than 2.75 m distance variation). While the implementation of such technologies is still pending, the integration of DTs can enable decision-makers to promote smart urban planning [64,114].

In summary, the deployment of digital twin technology in bus transport systems has led to measurable improvements in operational efficiency and maintenance practices. The DT case studies discussed were selected based on their relevance to practical and real-world implementations. By leveraging real-time data and predictive analytics, DTs can enhance service delivery and contribute to a passenger-friendly urban transportation system.

5. Discussion

Digital twin technology has been widely used in industries like manufacturing, maritime operations, and healthcare, but its potential in public transportation remains underexplored in urban and non-urban settings. The integration of DT technology into public bus transport systems presents significant impacts by improving operational efficiency and operations and reducing delays. As demonstrated by several studies, DTs can use real-time data analysis and predictive maintenance to predict traffic patterns, allowing for better planning of bus routes and schedules. For instance, a study in Brazil demonstrated a DT's ability to improve the traffic flow and reduce delays at intersections by simulating and predicting traffic situations [102]. Additionally, the implementation of DTs can assist in smarter decision-making to improve efficiency and scalability based on real-time transit demand [46]. Moreover, predictive maintenance also enables the early identification of issues within bus systems, ensuring a longer lifespan and reduced operational costs [34,62,100].

While the applications of DTs in public transit systems are considerable, there are certain gaps in the research that require attention. One of the identified gaps is that research has focused on short-term impacts without considering the long-term sustainability of these improvements. There were only a few studies that discussed how the benefits of DTs continue to scale over time. Additionally, there is a lack of research integrating diverse data sources such as passenger counts, vehicle health data, and environmental factors like weather into a single framework. The integrated system could be more beneficial in providing a more comprehensive approach to decision-making [49,102].

There is also a lack of studies that incorporate the scalability and adaptability of DTs across multiple geographical locations. While several studies focus on the successful implementation of DTs in well-developed urban areas, there is lack of research in smaller towns or rural areas. The latter places could have different traffic patterns and problems, less reliable infrastructure, or fewer resources [49]. Hence, it is important to understand how DTs could be adapted for non-urban areas [34].

Another concern is that there are only a few studies that address the risks and mitigation strategies to handle real-time and sensitive data. More research is required to strengthen cybersecurity measures and alleviate data privacy concerns that have been mentioned in different studies [35,63]. There is also a lack of comprehensive long-term economic feasibility of implementing DT technologies in bus transit systems. Although DTs improve bus operations, there are not many studies that have analyzed the costs involved. Transit authorities need to understand the potential economic benefits and trade-offs that will be essential to consider the adoption of DT technologies. Another crucial gap is the social equity implications of DTs in public transportation. There is not much research showing how DTs can help reduce transportation disparities, especially for underserved communities and regions with limited access to public transit services [34,49].

While DT technology shows multiple benefits for bus transit systems, it is important to address the gaps to standardize adoption and unlock its full potential. More research is required to understand the long-term sustainability of DTs' performance. There is also research required to integrate multiple data sources in order to determine the economic and social equity implications and feasibility in different geographic locations. Future studies should also explore standard ways to protect data privacy and cybersecurity without losing the benefits of real-time analysis. When these gaps are addressed, DTs can play an even more significant role in public transportation system operations, benefiting both passengers and operators.

6. Conclusions

Digital twin technology has immense potential to revolutionize public transportation systems addressing basic issues like inefficiency, safety, and a poor passenger experience. By creating dynamic and real-time digital versions of physical transit assets, DTs can enable predictive maintenance, the efficient allocation of resources, and data-driven decisions, all of which create resilient and adaptive transportation networks.

However, there are some obstacles that exist to the adoption of DTs in public transit. Some of the issues include the integration of the existing infrastructure into advanced technologies, data security and privacy, and interoperability between different systems. This paper has discussed some important methodologies, which could potentially shape the future of DT applications. These architectures are modular and micro-service-based, and they are easy to scale and adapt by organizations and policymakers. The integration of blockchain further enhances security and trust in data, while standardized communication protocols allow interoperability across different platforms. These approaches not only overcome the limitation of DT technologies but also provide a roadmap toward designing robust DT systems.

These findings point to the need for concurrent and sustained collaboration among research, industry, and government entities to meet DT-related challenges. Policymakers should support regulatory frameworks, while researchers should provide sustained innovation and refinement of methodologies. Similarly, workforce training and education also require investments to ensure that professionals can efficiently design, implement, and manage DT solutions. Training programs focused on data privacy, cybersecurity, and interoperability can help professionals efficiently design, implement, and manage DT solutions. Likewise, public education initiatives can also build trust in DT applications by increasing user awareness. This can be very beneficial for authorities if they are able to effectively incorporate feedback from the public to further improve transit operations and passenger experiences.

Full integration of DTs into public transit systems can be accomplished if there is a balanced comprehensive approach among technological developments, organizational aspects, and policy considerations. DTs can improve the efficiency, sustainability, and passenger satisfaction of transit only by fostering collaboration within the ecosystem and maintaining a focus on user-centric design. As cities continue to grow and evolve, DT technology can meet the increasing demands of urban mobility, allowing smarter and more connected communities.

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